

DIABETES PREDICTOR

COURSE TITLE: SOCIAL NETWORK ANALYSIS
CS5127



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INTRODUCTION

The main objective of this project is to develop a Machine Learning (ML)-based system for predicting post-pregnancy diabetes using significant medical attributes and patient health data. Logistic Regression (LR) is used to analyze and identify the key risk factors associated with diabetes. The model studies the relationship between various health parameters and predicts the likelihood of diabetes occurrence, enabling early diagnosis and better healthcare management. This predictive approach can assist healthcare professionals in taking timely preventive measures and improving maternal health outcomes.

LITERATURE REVIEW

Several “Diabetes Predictor” models have already been developed using different Machine Learning techniques. However, many existing systems still face challenges in achieving higher prediction accuracy and reliability. In healthcare-related applications, even a small improvement in accuracy can make a significant difference, as medical predictions directly impact patient health and treatment decisions. Therefore, achieving a highly accurate and dependable prediction model is extremely important. Our proposed model focuses on improving prediction performance by using optimized algorithms and better data analysis techniques. The system aims to provide more reliable results compared to many existing models, thereby supporting early diagnosis and effective preventive healthcare.

```
DT = DecisionTreeClassifier()
DT.fit(X_train, Y_train)
Y_pred = DT.predict(X_test)
print("Accuracy ", DT.score(X_test, Y_test)*100)
sns.set(font_scale=1.5)
cm=confusion_matrix(Y_pred,Y_test)
sns.heatmap(cm,annot=True)
```

Accuracy 70.77922077922078

```
LR=LogisticRegression()
LR.fit(X_train,Y_train)
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```

Accuracy 35.714285714285715



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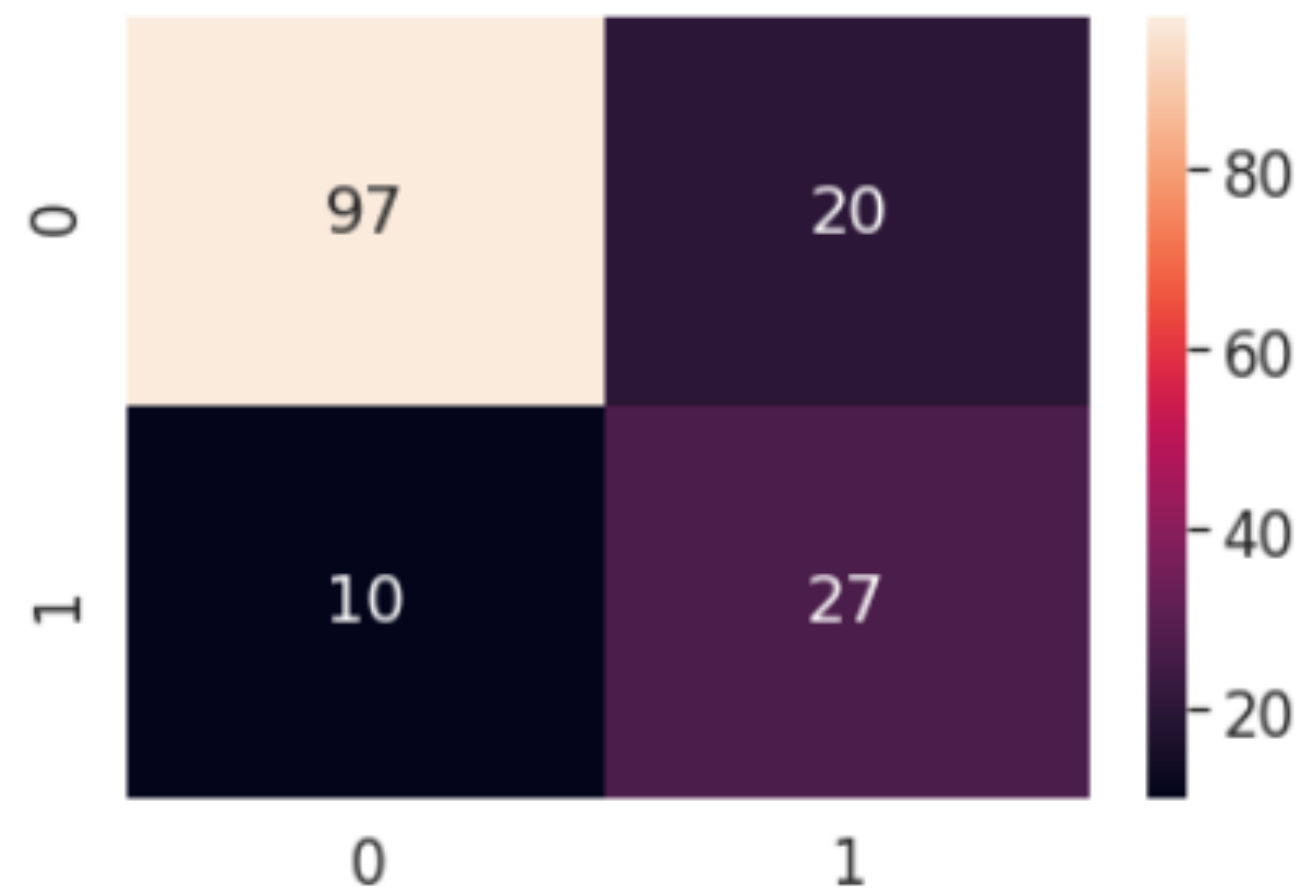
A model for early prediction of diabetes prediction of diabetes. The ANN technique provided a best accuracy of 75.7%, and may be useful to assist medical professionals with treatment decisions.

PROPOSED MODEL

```
LR=LogisticRegression()  
LR.fit(X_train,Y_train)  
Y_pred=LR.predict(X_test)  
print("Accuracy",LR.score(X_test,Y_test)*100)  
sns.set(font_scale=1.5)  
cm=confusion_matrix(Y_pred,Y_test)  
sns.heatmap(cm,annot=True)
```

Accuracy 80.51948051948052

<matplotlib.axes._subplots.AxesSubplot at 0x7f651435fe50>



Diabetes Checkup

Training Data Stats

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI
count	768.0000	768.0000	768.0000	768.0000	768.0000	768.0000
mean	3.8451	120.8945	69.1055	20.5365	79.7995	31.9926
std	3.3696	31.9726	19.3558	15.9522	115.2440	7.8842
min	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
25%	1.0000	99.0000	62.0000	0.0000	0.0000	27.3000
50%	3.0000	117.0000	72.0000	23.0000	30.5000	32.0000
75%	6.0000	140.2500	80.0000	32.0000	127.2500	36.6000
max	17.0000	199.0000	122.0000	99.0000	846.0000	67.1000

Patient Data

	pregnancies	glucose	bp	skinthickness	insulin	bmi	dpf	age
0	7	161	54	77	193	25	1.0300	33

EXPERIMENTAL SET-UP

Pandas



matplotlib



seaborn



Streamlit



NumPy

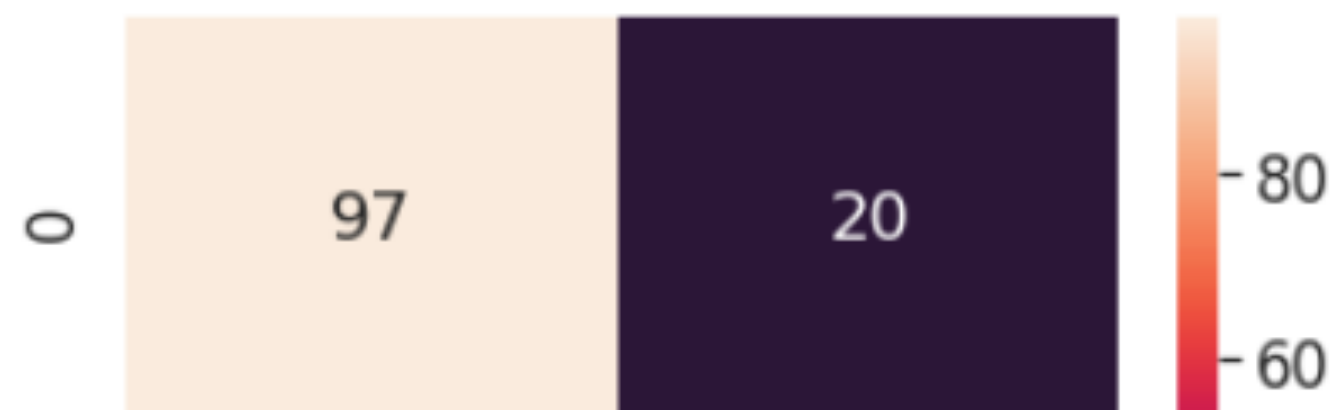


LIMITATION

Like every project, this system also has certain limitations. The primary limitation is the accuracy of the prediction model, which depends heavily on the quality and consistency of the dataset used for training. Since the available dataset contains some inconsistencies and variations, the model's performance may not always achieve optimal accuracy. Efforts are currently being made to improve data consistency and enhance the overall accuracy and reliability of the system through better preprocessing and model optimization techniques.

```
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LR.fit(X_train,Y_train)  
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print("Accuracy",LR.score(X_test,Y_test)*100)  
sns.set(font_scale=1.5)  
cm=confusion_matrix(Y_pred,Y_test)  
sns.heatmap(cm,annot=True)
```

```
Accuracy 80.51948051948052  
<matplotlib.axes._subplots.AxesSubplot at 0x7f651435fe50>
```



CONCLUSION AND FUTURE SCOPE

In conclusion, the “Diabetes Predictor” system successfully demonstrates the use of Machine Learning techniques in predicting post-pregnancy diabetes at an early stage. The project aims to assist individuals and healthcare professionals by providing a simple, efficient, and reliable method for identifying diabetes risk using medical data and predictive analysis. Early prediction can help in taking preventive measures and reducing the chances of severe health complications.

In the future, the project can be further enhanced by improving model accuracy and integrating advanced non-invasive technologies for diabetes detection. This could reduce the dependence on painful traditional methods such as repeated needle-based blood sugar testing and provide users with a more comfortable and accessible health monitoring solution.

Your Report:

You are Diabetic

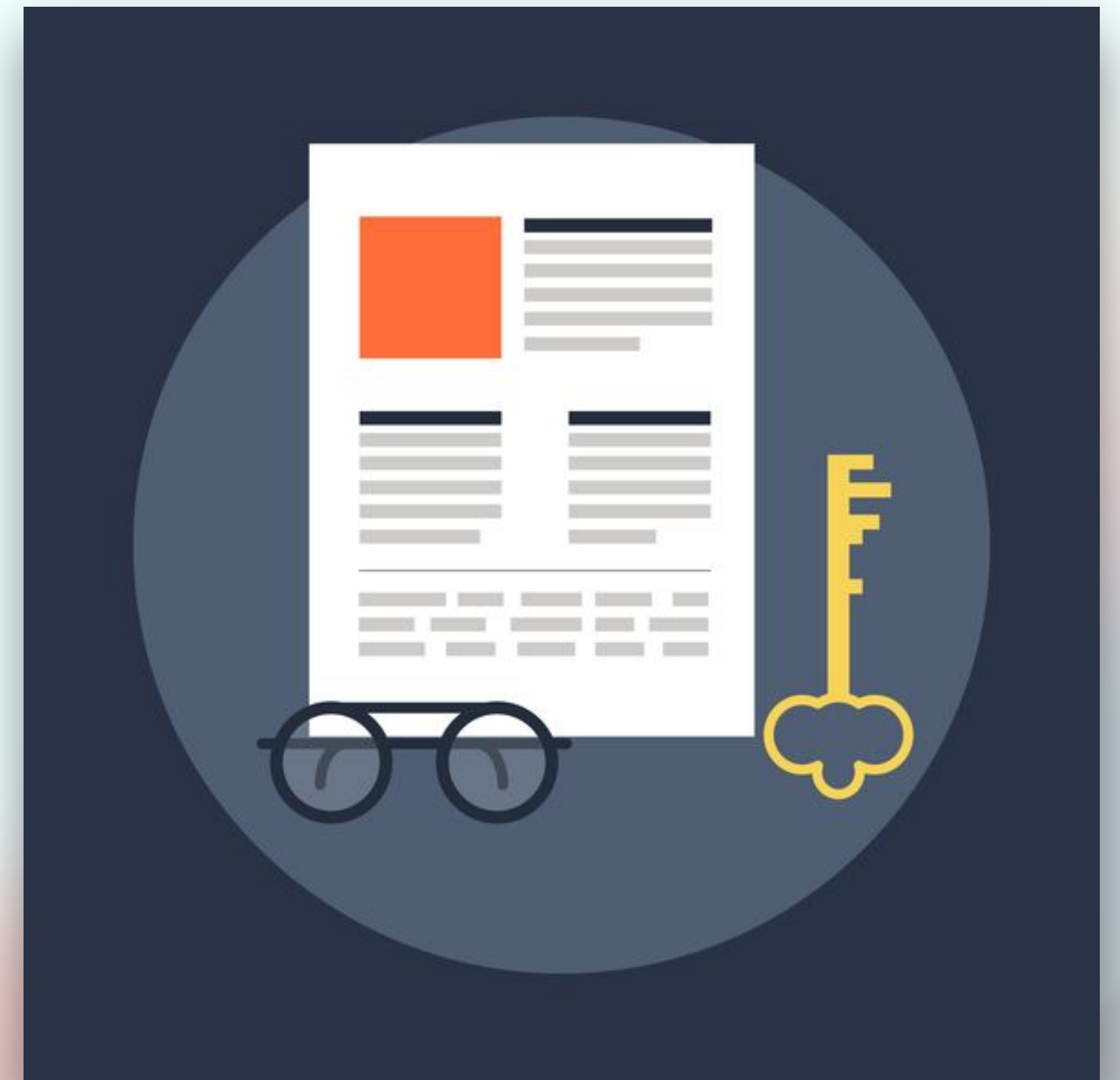
Accuracy:

82.46753246753246%



REFERENCES

- ☆ Google Article
- ☆ W3 School
- ☆ Kaggle



Thank
you!